

Muhammad Zain

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EDUCATION

University of Calgary

Bachelor of Science in Software Engineering

Calgary, AB

September 2022 - April 2027

- **Relevant Coursework:** Software Architecture, Algorithmic Design and Analysis, Concurrent and Distributed Systems, Computational Data Management, Software Performance Evaluation, Software Testing and Quality

SKILLS

- **Programming Languages:** Python, Java, C++, JavaScript, TypeScript, SQL, HTML, CSS
- **Backend & APIs:** Flask, REST APIs, JSON, MySQL Connection Pooling, Gemini API, LLM Integration
- **Frontend & UI:** React, Three.js, WebGL, Vite, Responsive Design
- **DevOps & Tooling:** Git, GitHub, GitHub Actions (CI/CD), Docker, Docker Compose, GCP (Cloud Run, Cloud SQL, GCS), MySQL, Agile

EXPERIENCE

Enactus Canada

Software Development Intern

Calgary, AB

September 2025 - May 2026

- Designed a spatiotemporal data pipeline aligning satellite, weather, & event datasets onto a canonical 30 km grid for weekly wildfire risk analysis.
- Implemented leakage-safe temporal validation & grid-based joins, preventing spatial & time contamination across multi-year machine learning experiments.
- Engineered feature workflows for extreme class imbalance, evaluating gradient-boosted models with PR-AUC & Recall@Top-K to prioritize high-risk regions.
- Standardized reproducible outputs including Parquet tables, model artifacts, & interactive risk maps for engineering handoff & operational review.

PROJECTS

Resume & Career Coach (RCC)

Python, Flask, React, PostgreSQL, Docker, Gemini LLM

Calgary, AB

January 2026 - April 2026

- Built a Dockerized Flask + React + PostgreSQL app with a hybrid token-overlap + LLM pipeline (Gemini).
- Integrated an OpenAI-compatible REST API with structured JSON prompting, regex parsing, & 80+ term domain allowlist.
- Designed an LLM-powered mock interview engine with adaptive questions & Web Speech API TTS/STT voice I/O.
- Deployed to production on Vercel, Render, & Neon, with a 3-service Docker Compose local setup & graceful AI-failure fallback.

ProfileHub Performance Evaluation

Python, Flask, MySQL, Locust, GCP, Pandas, SciPy

Calgary, AB

January 2026 - April 2026

- Deployed a Flask + React + MySQL service on Google Cloud Run with Cloud SQL & GCS via Gunicorn.
- Designed a 72-run factorial Locust experiment (workloads × sizes × concurrency × reps) with headless orchestration.
- Built a Python statistical pipeline computing 95% confidence intervals (Student's t) for latency & throughput.
- Diagnosed MySQL connection-pool leaks under load, improving service reliability under concurrent simulations.

CityX Subway Display System

Java, Swing, MySQL, Multithreading, REST APIs

Calgary, AB

June 2025 - July 2025

- Engineered a real-time system in clean MVC architecture with multithreaded pipelines for concurrent updates.
- Built a resilient REST integration layer with timeouts, 3-attempt retry/backoff, & caching for 90% reliability under flaky upstream APIs.
- Designed an interactive Swing UI with animated map transitions & fade/scale ad rotations.
- Wrote a 58-test JUnit suite (100% model coverage) covering edge cases, CSV parsing, & failure paths.

Flight Operations Manager

C++, Qt, STL Containers, SQLite, CMake

Calgary, AB

February 2025 - March 2025

- Developed a cross-platform desktop application managing 500+ flight, passenger, & schedule records.
- Architected an OOP domain model with safe memory ownership, applying Singleton, Observer, Factory, Strategy, & Command patterns.
- Designed a pluggable storage layer behind a Database interface with two backends (SQLite & JSON/CSV).
- Built with CMake for cross-platform compilation on Windows, macOS, & Linux.